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lab 10:

```
data class Product(
```

```
    val name: String,
```

```
    val category: String,
```

```
    val price: Double,
```

```
    val stock: Int
```

```
)
```

```
fun applyDiscount(
```

```
    products: List<Product>,
```

```
    discountFunction: (Double) -> Double
```

```
): List<Product> {
```

```
    return products.map { product ->
```

```
        product.copy(price = discountFunction(product.price))
```

```
    }
```

```
}
```

```
fun filterProducts(
```

```
    products: List<Product>,
```

```
    condition: (Product) -> Boolean
```

```
): List<Product> {
```

```
    return products.filter(condition)
```

```
}
```

```
fun main() {  
    val products = listOf(  
        Product("Laptop", "Electronics", 100000.0, 5),  
        Product("Phone", "Electronics", 50000.0, 15),  
        Product("Chair", "Furniture", 2000.0, 25),  
        Product("Table", "Furniture", 3000.0, 5),  
        Product("Pen", "Stationery", 50.0, 100)  
    )  
  
    val discountedProducts = applyDiscount(products) { price -> price * 0.9 }  
  
    println("--- Discounted Product List ---")  
    for (p in discountedProducts) {  
        println("Name: ${p.name} | Category: ${p.category} | Price: ${p.price} | Stock:  
${p.stock}")  
    }  
  
    val lowStock = filterProducts(discountedProducts) { it.stock < 10 }  
    println()  
    println("--- Low Stock Products (<10) ---")  
    for (p in lowStock) {  
        println("${p.name} (${p.stock})")  
    }  
  
    val premiumProducts = filterProducts(discountedProducts) { it.price > 1000 }  
    println()  
    println("--- Premium Products (>1000) ---")  
    for (p in premiumProducts) {
```

```
        println("${p.name} (${p.price})")
    }
}
```